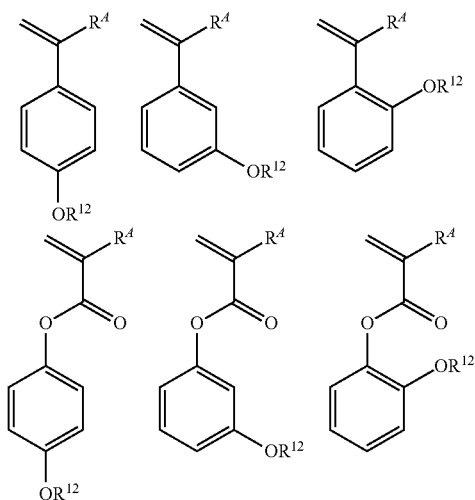
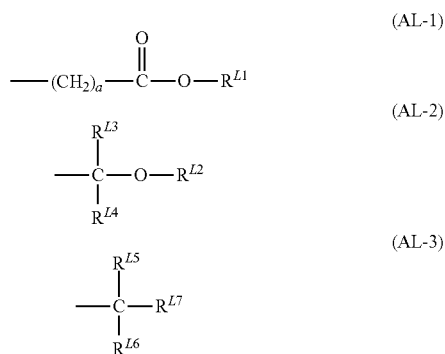




[0065] The acid labile groups represented by R^{L1} and R^{L2} in the recurring units (a1) and (a2) may be selected from a variety of such groups, for example, those groups described in JP-A 2013-080033 (U.S. Pat. No. 8,574,817) and JP-A 2013-083821 (U.S. Pat. No. 8,846,303).

[0066] Typical of the acid labile group are groups of the following formulae (AL-1) to (AL-3).



[0067] In formulae (AL-1) and (AL-2), R^{L1} and R^{L2} are each independently a C_1 - C_{40} monovalent hydrocarbon group which may contain a heteroatom such as oxygen, sulfur, nitrogen or fluorine. The monovalent hydrocarbon groups may be straight, branched or cyclic, with alkyl groups of 1 to 40 carbon atoms, especially 1 to 20 carbon atoms being preferred. In formula (AL-1), "a" is an integer of 0 to 10, especially 1 to 5.

[0068] In formula (AL-2), R^{L3} and R^{L4} are each independently hydrogen or a C_1 - C_{20} monovalent hydrocarbon group which may contain a heteroatom such as oxygen, sulfur, nitrogen or fluorine. The monovalent hydrocarbon groups may be straight, branched or cyclic, with C_1 - C_{20} alkyl groups being preferred. Any two of R^{L2} , R^{L3} and R^{L4} may bond together to form a ring with the carbon atom or carbon and oxygen atoms to which they are attached. The ring contains 3 to 20 carbon atoms, preferably 4 to 16 carbon atoms, and is typically alicyclic.

[0069] In formula (AL-3), R^{L5} , R^{L6} and R^{L7} are each independently a C_1 - C_{20} monovalent hydrocarbon group which may contain a heteroatom such as oxygen, sulfur, nitrogen or fluorine. The monovalent hydrocarbon groups may be straight, branched or cyclic, with C_1 - C_{20} alkyl groups being preferred. Any two of R^{L5} , R^{L6} and R^{L7} may bond together to form a ring with the carbon atom to which they are attached. The ring contains 3 to 20 carbon atoms, preferably 4 to 16 carbon atoms and is typically alicyclic.

[0070] The base polymer may further comprise recurring units (b) having a phenolic hydroxyl group as an adhesive group. Examples of suitable monomers from which recurring units (b) are derived are given below, but not limited thereto. Herein R^4 is as defined above.

